

Exercise_2

September 8, 2026

0.1 Compound Interest Rate

John deposited an initial amount of **Rs. 3000** in his account. After **3 years**, the balance reached **Rs. 3335.80** due to compound interest. What is the interest rate per annum?

```
[4]: # A (in CI) = P (1 + R/100)^T
import math

p = 3000
t = 3
a = 3335.8

r = (math.pow(a/p, (1/t)) - 1) * 100

print(f"The rate is = {r:.2f}")
```

The rate is = 3.60

0.2 Area of a Triangle

The three sides of a triangle are **3 m**, **4 m**, and **6 m**. Calculate its area using Heron's formula.

```
[5]: a = 3
b = 4
c = 6

s = (a+b+c) / 2

ar = (s * (s-a) * (s-b) * (s-c)) ** 0.5

print("Area is = ", round(ar, 2))
```

Area is = 5.33

0.3 Number Guessing Game

Generate a random number between **1 and 100** and let the user guess it. After each guess, say whether the guess is greater than, equal to, or smaller than the answer. End the game when the guess is correct.

```
[6]: import random

no = random.randint(1, 100)
count = 0

while True:
    guess = int(input("Guess a number between 1 and 100: "))
    count += 1
    if guess > no:
        print("Your guess is greater than the no. Try Again. \n")
    elif guess < no:
        print("Your guess is smaller than the no. Try Again. \n")
    else:
        print("Correct! Game over.")
        break

print(f"\n You guessed the no. {no} in {count} guesses. \n")
```

Your guess is smaller than the no. Try Again.

Your guess is smaller than the no. Try Again.

Your guess is smaller than the no. Try Again.

Your guess is smaller than the no. Try Again.

Your guess is smaller than the no. Try Again.

Your guess is smaller than the no. Try Again.

Correct! Game over.

You guessed the no. 99 in 7 guesses.

Ask the user for two integers, a and b. Print a if $a > b$, ab if $a == b$, and b if $a < b$.

```
[7]: a = int(input("Enter the first no: "))
b = int(input("Enter the second no: "))

if a > b:
    print(a)
elif a == b:
    print(a*b)
else:
    print(b)
```

Print the string Hello 123 times, with each occurrence on its own line.

```
[8]: for i in range(123):  
      print("Hello")
```

[illegible]

[illegible]

[illegible]

Ask the user for three integers, `a`, `b`, and `c`, and print the largest value. If the largest value occurs more than once, print `Tie`.

```
[9]: a = int(input("Enter a: "))
      b = int(input("Enter b: "))
      c = int(input("Enter c: "))

      print(max(a, b, c))

      if (a == b and a >= c) or (a == c and a >= b) or (b == c and b >= a) or (a == b
          and b == c):
          print("Tie")
```